

Pause & Protect: A Real-Time Scam Warning for K PLUS Transfers

Track 2: Data Science & Intelligence | KBTG Kampus Hackathon 2026

The Problem

First-jobbers are the K PLUS users most exposed to scams. They have only just started managing their own money, they do not yet recognise tactics like fake job offers, investment schemes, bank-impersonation SMS, and QR-code swaps, and they are easily rushed by a message demanding an urgent transfer. The protections K PLUS offers today, such as MFA and facial recognition, confirm who is making the transfer, not whether the transfer itself makes sense. Fraud can only be reported after the money has gone, by which point it has usually passed through a mule account and cannot be recovered.

The Solution

We propose a risk score shown at the moment a user is about to confirm a transfer, presented as a short question in plain language rather than a hard block. For example: *"You have not sent money to this account before, and the amount is much larger than usual. Is this a job offer, or an urgent request from someone you have not met?"* The score uses signals K PLUS already holds: whether the recipient is new to the user, how the amount compares with the user's own history, the time of day, recent transfer frequency, and whether the recipient has appeared in earlier fraud reports. A small, explainable model turns these into a low, medium, or high rating.

- Medium and high risk trigger a specific warning with "Proceed anyway" and "Cancel".
- Even a low-risk transfer to a known recipient shows a brief confirmation screen first, since a mistyped amount is a risk whatever the account.

Why This Is Different for Thailand and First-Jobbers

Differentiation. Real-time "confirm or decline" alerts are not new; banks such as ICICI, HDFC, and Standard Chartered already run versions of them, so the idea itself is not the contribution. What we believe is different is the categories and how we plan to treat new users. Our scam labels, investment scams, job-offer scams, QR-clone and bank-impersonation attempts, mule-account patterns, follow categories the Bank of Thailand tracks through its 2023 mule-account measures, and are weighted toward PromptPay's QR-based transfers rather than the card and wire patterns most alert systems were built around. Most transfer-risk systems also need months of a customer's own history, which a first-jobber only weeks into using K PLUS does not yet have. Our design would blend thin personal history with a peer-group baseline from similarly aged, similarly tenured customers until enough history builds up, though this part is designed, not yet built into the prototype, which still assumes an established baseline.

Reducing false alarms. A system that warns too often gets ignored, so we treat false alarms as seriously as missed scams, in three ways. Only medium and high risk show a visible warning, so most transfers pass with just the universal confirmation step. A warning is a pause, not a block, so a false alarm costs a few seconds of doubt, not a stopped payment. And every outcome, a user proceeding or cancelling, and whether a transfer is later confirmed as fraud, is meant to become labelled data that loosens the model on false alarms and tightens it on missed scams. That feedback loop is designed but not yet running, since we have no real outcome data yet.

Value to KBTG. This is built for people in their first one to three years of work, the group the hackathon brief names and the one with the least banking experience to fall back on when a scam is convincing. It cuts fraud losses in the customer group where each loss does the most reputational harm, and under the shared-responsibility decree, a documented pre-transfer warning is a record that the bank acted responsibly, which carries legal and commercial weight beyond goodwill.

Prototype and Results

We have built a working demo of the transfer flow.

- The risk model is a gradient-boosted classifier, trained and tested on synthetic data built to reproduce fraud patterns from the Bank of Thailand's mule-account reporting, since we lack access to real KBank transaction data. On this synthetic set it reaches 94% overall accuracy and 86% recall on the scam class, which **shows the modelling approach works**, not what accuracy K PLUS would see in production; that would require training on K PLUS's own transaction and fraud-report history.
- A risk gauge updates as the amount or time of the transfer changes, so the assessment responds to what the user enters.
- Each flagged transfer is labelled with the scam category it most resembles, such as "investment scam" or "late-night transfer to a new recipient".
- A confirmation step runs before every transfer, whatever the risk level or recipient.